MATH0005 Algebra 1

Year:	2026–2027
Code:	MATH0005
Level:	4 (UG)
Normal student group(s):	UG: Y1 Mathematics programmes
Value:	15 credits (7.5 ECTS credits)
Term:	1
Assessment:	75% examination, 15% coursework, 10% departmental test(s). The departmental test is a midsessional exam in January. In order to pass the module you must have at least 40% for both the examination mark and the final weighted mark.
Normal Pre-requisites:	A* in A-level Mathematics and Further Mathematics
Lecturer:	Dr M Towers

Course Description and Objectives

This course is an introduction to linear algebra: the theory and methods for vectors and matrices. It is an essential foundational topic for all branches of pure and applied mathematics, and any other subject where mathematics is used. Problems in linear algebra can be expressed via matrices, by systems of linear equations, or by geometry, and we will see how to use all these points of view. We will learn the core concepts of subspace and dimension, the rank of a matrix, and the powerful rank-nullity theorem. We will discover how to characterise matrices abstractly as linear maps, and various basic techniques for working with matrices including row reduction, determinants, and eigenvalues.

Recommended Texts

You are not required to buy any books for this course, but you may find the following helpful:

– DA Towers, *Guide to Linear Algebra* (Palgrave)

– AO Morris, *Linear Algebra* (Chapman and Hall)

Detailed Syllabus

Matrix manipulation: Matrices and vectors over $\mathbb{R}$ and $\mathbb{C}$. Addition, scalar multiplication, transpose, trace, matrix multiplication, invertibility. Linear systems and matrix equations. Row operations, row reduced echelon form, elementary matrices. Use of RREF for invertibility and inverses.

Geometry and subspaces: Lines and planes in $\mathbb{R}^2$ and $\mathbb{R}^3$. Finding intersections. Dot product, lengths and angles. Definition of a subspace of a space of column vectors. Linear independence, spanning, their geometric meaning. Bases. Steinitz exchange lemma, proof that every subspace has a dimension. Extending a basis. Affine subspaces.

Linear maps and rank-nullity: Linear maps between spaces of column vectors and matrices for linear maps. Matrix multiplication as composition. Kernel, image, rank. Rank-nullity. Relationship between rank and surjectivity, injectivity, and invertiblity. A matrix is invertible if and only if its columns form a basis.

Determinants: Determinants for 2×2 and 3×3 matrices. Interpretation as area/volume. Determinants for larger matrices (sketched). Multiplicativity. Determinants and invertibility. Determinants of diagonal and triangular matrices.

Changing basis: Linear changes of co-ordinates. Expressing a vector in different bases. The change of basis matrix. Conjugation as a co-ordinate change and its relationship to the representation of a linear map in a new basis. Invariance of determinant and trace.

Eigenvectors and diagonalization: Eigenvectors, eigenvalues, the characteristic polynomial. Fundamental theorem of algebra (statement only). Diagonalization over $\mathbb{R}$ and $\mathbb{C}$.

Orthogonality: Orthogonal matrices, orthonormal bases. Gram-Schmidt. A symmetric matrix is diagonalizable.

September 2026